Mike Ichikawa

DATA SCIENTIST · ML ENGINEER · DATA ENGINEER

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SUMMARY

MS Mathematics (CCNY, 2014) and BS Mechanical Engineering (UC Berkeley). Three years in semiconductor Physical Design at Intel; nine years teaching undergraduate mathematics at Santa Rosa Junior College. Over the past eighteen months, a self-directed transition into data science and ML engineering — sixteen production-quality portfolio projects built from scratch across time series forecasting, anomaly detection, NLP, cloud ETL, ML serving, a Databricks medallion lakehouse, and C++ performance extensions via pybind11, plus a complete five-repo paper trading system: live market data ingestion through dual-path signal generation, parameter-optimizing backtesting, and a live Streamlit oversight dashboard.

TECHNICAL SKILLS

Languages	Python · SQL · C++ · R · TypeScript · Node.js
ML / Statistics	scikit-learn · Prophet · PyTorch · XGBoost · LightGBM · scipy · statsmodels · Bayesian inference
LLM / NLP	Anthropic API · Azure OpenAI · Microsoft Graph · MSAL · HuggingFace · FinBERT · prompt engineering
Data Engineering	PostgreSQL · Databricks · Delta Lake · medallion · dbt · SQLAlchemy · pandas
Cloud / Infra	AWS (S3 · Lambda · DynamoDB · CloudWatch) · Docker · FastAPI · Redis · Pydantic v2 · boto3
Tools	Git · pytest · Vitest · Jupyter · Streamlit · Docker Compose

PORTFOLIO PROJECTS — full portfolio at mtichikawa.github.io

Trading System Arc
Live Demo · github.com/mtichikawa — 5 repos

Five interconnected repositories: live OHLCV pipeline (ccxt/Kraken + PostgreSQL) → candlestick chart generation (mplfinance) → dual-path signal engine (EMA/RSI/MACD/BB + local FinBERT sentiment, 0.6/0.4 fusion) → parameter-optimizing backtester (Sharpe/Sortino/drawdown, staged grid search, T3 feedback loop) → Streamlit multi-page oversight dashboard with live signal display, equity curve, and parameter review. 174/174 tests across T2–T5.

ccxt · PostgreSQL · mplfinance · FinBERT · pandas · NumPy · Streamlit · Plotly · pytest

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Databricks Medallion Lakehouse
github.com/mtichikawa/databricks-lakehouse

Medallion lakehouse (bronze → silver → gold) on 10M NYC taxi rows with Delta Lake. Append-only bronze with provenance metadata; silver types, dedupes, and quarantines invalid rows rather than dropping them; gold serves query-specific aggregates. Row-level data quality gates between layers halt the pipeline on schema drift or null-rate breaches. 95 tests across bronze/silver/gold/quality/pipeline. Runs locally via deltalake; deployable to Databricks Community Edition.

Delta Lake · medallion architecture · pandas · pyarrow · data quality gates · pytest

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Real-Time Anomaly Detection
github.com/mtichikawa/anomaly-detection

Streaming ensemble detector: Isolation Forest, LSTM autoencoder, and Z-score with 2-of-3 majority voting. Dockerized FastAPI REST API with health checks and configurable detector selection. Live interactive demo on portfolio site.

Docker · FastAPI · scikit-learn · PyTorch · Redis · pytest

Cloud ETL Pipeline
github.com/mtichikawa/cloud-etl-pipeline

AWS-native ETL: tenacity-based REST API ingestion with exponential backoff → Parquet/Snappy storage on S3 with Hive date partitioning → Lambda validation → DynamoDB run logging for failure traceability. Tests mock all AWS calls with moto.

AWS S3 · Lambda · DynamoDB · boto3 · pandas · Apache Parquet · moto

Dockerized ML API

github.com/mtichikawa/dockerized-ml-api

Production ML serving: async thread pool inference, Redis TTL caching (cache hit skips model entirely), Pydantic v2 request validation, async job queue with polling endpoints, Docker Compose deployment. 32 tests.

FastAPI · Docker · Redis · asyncio · Pydantic v2 · scikit-learn

SQL Analytics Pipeline

github.com/mtichikawa/sql-analytics-pipeline

Three-layer dbt-style pipeline (staging → clean → marts) on NYC Airbnb data. Window functions, CTEs, SQLAlchemy ORM with swappable SQLite/PostgreSQL backend. Layered transform pattern enforces raw data immutability.

PostgreSQL · SQLAlchemy · dbt-style transforms · SQLite · pandas

LLM Data Analysis Assistant

github.com/mtichikawa/llm-data-assistant

Two-path query routing: rule-based fast path handles ~70% of queries (correlations, aggregations, group-bys) with zero API cost. Queries below confidence threshold escalate to Anthropic API with DataFrame context injection and multi-turn conversation history.

Anthropic API · pandas · hybrid routing · multi-turn conversation

Microclaw (microclaw.app)

AI assistant for Microsoft 365 inside Microsoft Teams. Transactable SaaS on the Microsoft Commercial Marketplace (submitted to AppSource, certification in review). Node.js backend with Azure OpenAI smart routing, Microsoft Graph API across 12 M365 services, delegated Microsoft Entra authentication, Fulfillment API v2 + Marketplace Metering Service API for billing (Microsoft as merchant of record), event-driven Graph webhooks, two-layer permission system, per-tenant isolation. 200+ tests (Vitest), TypeScript strict mode.

TypeScript · Node.js · Azure OpenAI · Microsoft Graph · Teams Bot Framework · Marketplace Fulfillment API v2 · Marketplace Metering Service · Vitest

Additional: Multi-Armed Bandit A/B Testing (Thompson Sampling · UCB1 · Bayesian inference) · Bias Detection in LLMs (ANOVA · Cohen's d) · Financial NLP Parser (SEC EDGAR · regex · sentiment lexicon) · GitHub Trend Forecaster (Prophet · PyTorch LSTM in progress)

EXPERIENCE

Founder & Engineer

2026 – present

Microclaw LLC · Portland, OR

- Architected and shipped a transactable SaaS AI assistant on the Microsoft Commercial Marketplace (submitted to AppSource, certification in review), serving Microsoft 365 users through Microsoft Teams with 54 tools spanning 12 M365 services
- Integrated Microsoft Commercial Marketplace Fulfillment API v2 and the Marketplace Metering Service API for transactable SaaS billing with per-message metered overage — Microsoft as merchant of record — across an 8-tier bracket pricing model (Solo through Team-100, $25 per seat, 1,500 messages per seat pooled across the team) with application-enforced seat caps
- Designed smart GPT-4o / GPT-4o-mini routing and keyword-based tool selection targeting ~65% inference cost and ~60% input token reductions at projected query mix
- Delivered 200+ unit tests (all passing), 29 iterative testing rounds, two-layer permission system (organization + app-level toggles), and per-tenant data isolation
- Conducted primary-source policy research on Microsoft Commercial Marketplace certification rules mid-implementation; identified that the planned third-party payment overage design would violate the merchant-of-record requirement and pivoted to the compliant Marketplace Metering Service path before submission
- Led end-to-end GTM: Oregon LLC formation, EIN, USPTO trademark filing, Microsoft Partner Center and D&B DUNS onboarding, brand identity, and three showcase videos for launch

Mathematics Faculty
2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught undergraduate mathematics for nine years: Algebra through Calculus III, Linear Algebra, Differential Equations, Statistics
- Designed and maintained curriculum assessment systems to track student outcomes across multiple cohorts
- Developed data-driven iterative improvements to course materials based on longitudinal performance analysis
- Routinely explained complex mathematical concepts clearly to students with varying preparation levels

Physical Design Engineer
2008 – 2011

Intel Corporation · Semiconductor R&D

- Microprocessor physical design on Intel Ivy Bridge (22nm) — timing closure, power analysis, design rule verification
- Built Python and Perl automation for physical verification workflows, processing millions of layout data points per run
- Identified and flagged yield-risk patterns in pre-tape-out statistical data, contributing to sign-off decisions

EDUCATION

MS Mathematics — The City College of New York — 2014

BS Mechanical Engineering — University of California, Berkeley — 2008